

Practice problems PMT 1022

1. Prove that sum of any rational and irrational is irrational.
2. Prove that product of two odd integers always odd.
3. Prove that if the product of two integers is odd then both integers must be odd.
4. Does the sum of two irrationals always irrational? Justify your answer.
5. Prove that product of a non-zero rational and irrational is irrational
6. Prove that $\sqrt{2}$ is irrational. Deduce that $\frac{\sqrt{2}}{1+\sqrt{2}}$ is irrational.
7. Prove that between any two rational numbers there is a rational number.
8. Prove that between a rational number and irrational number, there is an irrational number.
9. State the Well Ordering principle
10. Prove Principle of mathematical induction, using the well ordering principle.
11. Is $\frac{22}{7}$ a rational number? Justify your answer.
12. Find the corresponding rational number for the following decimal expansion.
0.114523232323... where 23 is repeating forever.
13. Does the following Decimal expansion represent a rational number? Justify your answer.
0.101001000100001000001 ... where this pattern continues forever.
14. Find the base 10 representation of $(4536)_7$
15. Prove that If the base 10 representation of a number is $a_m a_{m-1} \dots a_2 a_1 a_0$, then
3 divides $a_m a_{m-1} \dots a_2 a_1 a_0$ if and only if 3 divides $a_m + a_{m-1} + \dots + a_2 + a_1 + a_0$.
16. Convert base 10 representation 1234 to a binary number.
17. State whether following statements are true or false. Justify your answer
 - (i) If $a|b$ and $b|a$ then $a = b$ or $b = -a$
 - (ii) If $a|(b+c)$ then $a|b$ or $a|c$
 - (iii) If $a \nmid b$ then $b \nmid a$
 - (iv) If $a|b$ and $c|d$ then $ac|bd$
18. Prove that an integer of the form $4k + 3$, cannot be written as sum of two perfect squares.
19. An integer of the form $4k + 2$ never a perfect square.
20. Let a, b, c, d, e are integers. Suppose $1|a, 2|ab, 3|abc, 4|abcd$, and $5|abcde$ Is that necessary to be $1.2.3.4.5|abcde$? Justify your answer.
21. Without calculating the exact value, find the last digit of each of the followings.
 - (i) 5002375^6
 - (ii) 711237^{19}
 - (iii) 700002^9
 - (iv) $2^{100} + 3^{100}$

22. Let a and b be integers, not both zero. Prove that a and b are relatively prime if and only if there exist integers x and y such that $ax + by = 1$.
23. Let a , b and c be integers. Prove that, If $a|c$ and $b|c$, with $\gcd(a,b) = 1$ then $ab|c$.
24. Let a , b and c be integers. Prove that, if $a|bc$ with $\gcd(a,b) = 1$ then $a|c$.
25. Prove that $\sqrt{5}$ is irrational.
26. Prove that $\sqrt{35}$ is irrational.
27. Use Euclidean Algorithm to find $x, y \in \mathbb{Z}$ such that $34x + 21y = 1$.
28. A house wife spends 1770 Rupees for purchasing mangoes and apples. She pays 31 Rupees for each mango and 21 Rupees for each apple. If she buys more mangoes than apples, how many mangoes and apples she buys?
29. Prove that there are infinitely many prime numbers.
30. Prove that any prime number greater than 3 is of the form $6k - 1$ or $6k + 1$ for some positive integer k . Deduce that if p is a prime and $p > 3$ then 24 is a factor of $p^2 - 1$.
31. Find the smallest positive integer d such that $7^d \equiv -1 \pmod{11}$.
32. Let p be a prime number and $p \geq 5$. Prove that $p^2 \equiv 1 \pmod{24}$.
33. Find the remainder when 7^{2026} is divided by 11.
34. Prove that If $a \equiv b \pmod{n}$, then $a + c \equiv b + c \pmod{n}$ and $ac \equiv bc \pmod{n}$.

